ZOO-501 Developmental Biology Mid term 2019

M.C.Qs

1. The birth defect study is called_____
2. The two stages of cell_____
3. Zp3 function is_____
4. Chilchodia larvae is found in_____
5. There are _____ strategies of commitment.
6. What is the is the regulative speciation_____
7. The human brain has _____ ganglia.
8. The sea urchin egg _____

Short Question

1. What is the process of primary neurulation (2)

Primary neurulation	Secondary neurulation
In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.	In **secondary neurulation**, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube.

Neurulation in fishes is exclusively secondary.

2. What is cavitation (2)

During a process called **cavitation**, the trophoblast cells secrete fluid into the morula to create a blastocoel. The inner cell mass is positioned on one side of the ring of trophoblast cells. The resulting structure, called the **blastocyst**, is another hallmark of mammalian cleavage.

3. Difference between sex and reproduction (2)

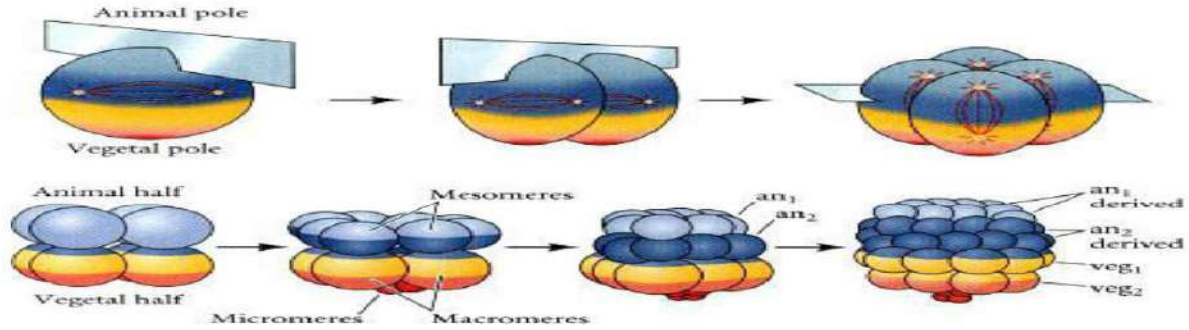
Sex and reproduction are two distinct and separable processes. **Reproduction** involves the creation of new individuals.

Sex	Reproduction
Sex involves the combining of genes from two different individuals into new arrangements.	**Reproduction** in the absence of sex is characteristic of organisms that reproduce by fission (i.e., splitting into two); there is no sorting of genes when an amoeba divides or when a hydra buds off cells to form a new colony.

Sex without reproduction is also common among unicellular organisms. Bacteria are able to transmit genes from one individual to another by means of sex pili. This transmission is separate from reproduction.

4. Difference between animal and vegetal pole (2)

The yolk-rich pole is referred to as the **vegetal pole**; the yolk concentration in the **animal pole** is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.



5. Protostomes detail (2)

Protostomes (Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location.

6. Function of MPF (3)

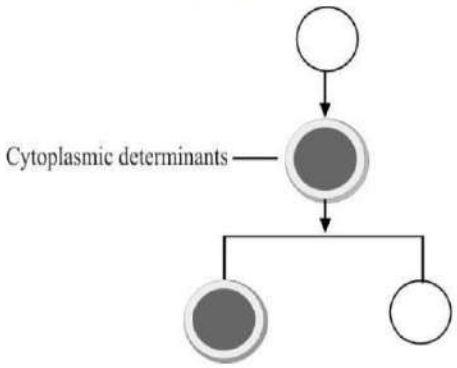
Maturation promoting factor (MPF) is a **cell** cycle checkpoint that regulates the passage of a **cell** from the G2 growth phase to the M phase. It is also known as the G2 checkpoint, and ensures that DNA replication during the S phase did not produce any mistakes.

7. What are the strategies for developmental commitment (3)

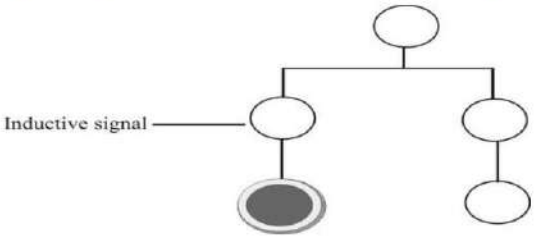
There appear to be two major strategies for establishing commitment and hence initiating the series of events that result in cell differentiation.

1. **The inheritance of cytoplasmic determinants:** Cytoplasmic determinants are the molecules in cytoplasm that can help to determine cell fate. The asymmetric distribution of cytoplasmic determinants indicates that the mechanism of differentiation is entirely intrinsic.
   For example, if a mother cell contains cytoplasmic determinants that is localized to one pole as the cell under goes division, that determinant will be inherited by only one of the daughters.
2. **The perception of external inductive signals:** The process where one cell or group of cells changes developmental fate of another is termed induction. It is extrinsic process that depends on the position of a cell in the embryo.
   Two identical cells can follow alternative fates if one is exposed to an external signal (often secreted by a different cell).

The inheritance of cytoplasmic determinants



The perception of external inductive signals



8. Difference between gray matter and white matter (3)

In spinal cord, the mantle zone, containing the neuronal cell bodies, is often referred to as the **Gray matter**; the axonal, marginal layer is often called the **White matter.**
After *neurulation*, the **gray matter** (mantle) gradually becomes a butterfly-shaped structure surrounded by **white matter**; and both become encased in connective tissue.

9. What is the differences between primary and secondary neurulation (3)

Primary neurulation	Secondary neurulation
In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.	In **secondary neurulation**, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube.

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Long Question

1. Functions of cell membrane.(5)

The cell membrane surrounds the cytoplasm of living cells, physically separating the intracellular components from the extracellular environment.

Functions:

i. The cell membrane also plays a role in anchoring the cytoskeleton to provide shape to the cell, and in attaching to the extracellular matrix and other cells to hold them together to form tissues.

ii. The cell membrane is selectively permeable and able to regulate what enters and exits the cell, thus facilitating the transport of materials needed for survival. The cell employs a number of transport mechanisms that involve biological membranes:

- ✓ Passive osmosis and diffusion
- ✓ Facilitated diffusion
- ✓ Endocytosis
- ✓ Exocytosis

2. Explain the process of gene regulation (5)

The regulation of gene expression can be accomplished at several levels:

Differential gene transcription:

It regulates that which of the nuclear genes are transcribed into RNA.

Selective nuclear RNA processing:

It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.

Selective messenger RNA translation:

It regulates that which of the mRNAs in the cytoplasm become translated into proteins.

Differential protein modification:

It regulates that which proteins are allowed to remain or function in the cell.

3. What factors cause the movement of gastrulation process (5)

The movements of gastrulation include invagination, involution, ingression, delamination, and epiboly.

This movement is not due to the active crawling of the blastomeres, however. Rather, the movement is provided by the autonomously expanding YSL "within" the animal pole yolk cytoplasm.The EVL is tightly joined to the YSL and is dragged along with it. The deep cells of the blastoderm then fill in the space between the **YSL** and the **EVL** as epiboly proceeds. This can be demonstrated by severing the attachments between the **YSL** and the **EVL.** When this is done, the **EVL** and deep cells spring back to the top of the yolk, while the **YSL** continues its expansion around the yolk cell. The expansion of the **YSL** depends on a network of microtubules in the **YSL**, and radiation or drugs that block the polymerization of tubulin inhibit epiboly.